

Uday Lunawat

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SUMMARY

Senior AI Platform Engineer with nearly 7 years of experience designing cloud-native infrastructure and internal developer platforms (IDP). Specialized in automating the lifecycle of AI systems using Infrastructure as Code (IaC) and CI/CD at enterprise scale across AWS and GCP. Proven track record of converting experimental GenAI initiatives into governed, production-grade enterprise systems. Deep experience in integrating FinOps accountability, Zero-Trust security, and agentic orchestration (LangGraph/RAG) into robust platforms that balance rapid innovation with operational reliability.

WORK EXPERIENCE

Senior AI Platform Engineer (GenAI, LLMOps, CI/CD) – Fractal Analytics, Pune

Nov 2024 – Present

Python, GCP, LangGraph, FastAPI, Jenkins, OIDC, SQLAlchemy, Docker

- Architected a centralized 'Agentic Factory', standardizing the lifecycle for 400+ AI agents to enable secure, multi-team adoption for a Fortune 5 U.S. Telecom.
- Unified fragmented pipelines into a modular CI/CD standard supporting 16,000+ deployments; reduced onboarding time to 90 seconds via automated infrastructure provisioning.
- Designed a stateful orchestration layer (LangGraph) supporting hierarchical reasoning, dynamic tool injection, and reliable traffic management, abstracting complexity for downstream developers.
- Engineered an automated cost-governance engine that detects inactive services, auto-reclaims quotas, and enforces strict budget guardrails on compute resources.
- Enforced a Zero-Trust security model using automated OIDC federation and granular RBAC, ensuring secrets isolation and enterprise compliance.
- Led a cross-functional team of 5 engineers, establishing design review standards and driving the technical roadmap for production AI maturity.

Senior Machine Learning Engineer – HCL, Bengaluru

Apr 2024 – Jun 2024

Python, GCP, MLflow, Seldon Core, TensorFlow, Label Studio, Flask

- Architected human-in-the-loop (HITL) evaluation frameworks for enterprise models, integrating human feedback loops for model validation.
- Developed scalable annotation pipelines using Label Studio and microservices on GCP, ensuring high-quality ground truth for retraining.
- Productionized ML inference services with Docker and Cloud Run, ensuring high availability and observability.

Machine Learning Engineer – Chugani, Bengaluru

Feb 2022 – May 2023

Python, AWS, MLflow, IaC, nbdev, Quantization

- Engineered a real-time liveness detection system, optimizing face recognition models via quantization to achieve 150ms latency.
- Modernized ML infrastructure by implementing MLflow for experiment tracking and Infrastructure as Code.
- Refactored legacy codebases into modular Python packages (nbdev), improving maintainability and collaboration.

Machine Learning Engineer – Yang, Bengaluru

Apr 2019 – Jan 2022

Python, Airflow, AWS, ETL, Tableau

- Built an automated AI audit system processing real-time factory data streams to improve quality control standards.
- Designed and deployed operational dashboards for 20+ vendor sites, saving over 200 manual hours per week.

- Orchestrated distributed data pipelines ensuring high availability for global stakeholders across 12 countries.

Systems Engineer Intern – Infosys, Mysuru

Oct 2018 – Mar 2019

Analytics Intern – TCS, Nagpur

Jan 2018 – Jul 2018

TECHNICAL SKILLS

Cloud & DevOps: AWS, GCP, Cloud Run, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

MLOps Platforms: MLflow, Seldon Core, FastAPI, Langfuse, Internal Developer Platforms (IDP)

Agentic AI & RAG: LangGraph, LangChain, Vector Databases (Chroma/FAISS), Function Calling, MCP (Model Context Protocol)

Generative AI Tooling: OpenAI/Gemini/Claude APIs, Claude Code, GitHub Copilot, Hugging Face, Prompt Engineering, Benchmarking

FinOps & Governance: Cost Attribution, Usage Telemetry, Budget Guardrails, RBAC

Languages & Libs: Python, SQL, Bash, Groovy, Pandas, NumPy, TensorFlow, PyTorch, Pydantic

OPEN SOURCE & PROJECTS

ALPR (Streamlit Hall of Fame) – License plate detection and OCR system deployed as Streamlit app |

PDF2Podcast – Converts PDF documents into podcast-style audio using generative AI |

RockReveal AI – Image classification system for rock identification using TensorFlow |

Idea Generation – Ideation tool using OpenAI API for generating innovation ideas